



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 5 • STANDARD 6.GR.1

Determine the Area of Parallelograms and Rhombuses

Lesson 5-1 • Teacher Copy

Name: Type your name

Date: Today's date

Class: Class period



TODAY'S OBJECTIVES

Content Objective: I can find the area of a parallelogram using base $\times$ height.

Language Objective: I can explain how I found the area using the words base, height, area, and formula.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Slanted side

Parallelogram

Parallel

Base

Height

Perpendicular

Area

Composite figure

Formula

Rhombus

Diagonal



Tap any word to see its meaning and a picture.

1

The leaning side of a parallelogram, which is NOT the height, is the

2

A quadrilateral with two pairs of parallel sides is a

3

Two sides that stay the same distance apart and never meet are

4

The side of a parallelogram you measure the height from at a 90° angle is the

5

The perpendicular distance between the base and the opposite side is the

6 Two lines that meet at a 90° square corner are .

7 The amount of flat space inside a two-dimensional figure is its .

8 A shape made of two or more basic shapes combined is a .

9 A math rule written with symbols, like $A = b \times h$, is a .

10 A parallelogram whose four sides are all the same length is a .

11 A segment from one corner to the opposite corner is a .

Write About the Math

How many tiles does the architect need?

¿Cuántos azulejos necesita el arquitecto?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Slanted side**
Lado inclinado

☐ **Parallelogram**
Paralelogramo

☐ **Parallel**
Paralelas

☐ **Base**
Base

☐ **Height**
Altura

Model: *We use the height because it is perpendicular to the base.* — Students connect 'height' to the perpendicular (straight-up) distance from base to top, and recognize the slanted side is longer and would overstate the area.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The courtyard's area is ____ sq ft because $A = \text{____} \times \text{____}$. She needs ____ tiles because ____ / 2 = ____.**

- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **My evidence is ____, so ____.**

Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Slanted side
2. **Notes 2:** Parallelogram
3. **Notes 3:** Parallel
4. **Notes 4:** Base
5. **Notes 5:** Height
6. **Notes 6:** Perpendicular
7. **Notes 7:** Area
8. **Notes 8:** Composite figure
9. **Notes 9:** Formula
10. **Notes 10:** Rhombus
11. **Notes 11:** Diagonal
12. **Watch for this mistake:** *The most common mistake in Area of Parallelograms is multiplying the base by the slanted side instead of the perpendicular height. For a parallelogram with a base of 16 ft, a perpendicular height of 9 ft, and a slanted side of 11 ft, students compute $16 \times 11 = 176$ square feet instead of the correct $16 \times 9 = 144$ square feet. The height must be the straight-up (perpendicular) distance between the base and its opposite side, not the length of the slanted side, even though the slanted side is often the number that 'looks' like it belongs in the formula.*
13. **Try It 1:** C. 40 sq ft — *Area = base $\times$ height = $8 \times 5 = 40$ sq ft. The 7 ft slanted side is not used; only the perpendicular height belongs in the formula.*
14. **Try It 2:** C. Equal: both are $10 \times 6 = 60$ sq ft because area depends on base and height, not the slant — *Area = base $\times$ height. Both have base 10 ft and height 6 ft, so both areas are $10 \times 6 = 60$ sq ft. Tilting the shape changes the slanted side but not the perpendicular height, so the area stays the same.*